

1 Solution — GIAM 6.5

Problem 4

1.1 Problem

Cantor's Snake builds a bijection $f : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0}$ by zig-zagging through the table whose cell at row $r \geq 1$ and column $c \geq 0$ holds c/r , listing each rational the *first* time it appears (i.e. when its representative is in lowest terms). Given $f(1) = 0/1$, $f(2) = 1/1$, $f(5) = 1/3$, compute:

- $f(26)$
- $f(30)$
- $f^{-1}(3/4)$
- $f^{-1}(6/7)$

1.2 Setup — How the Snake Walks

Cells are indexed (r, c) with $r \geq 1$ and $c \geq 0$; the cell holds the rational c/r . The **antidiagonal of sum s** is

$$D_s = \{ (1, s-1), (2, s-2), \dots, (s, 0) \}.$$

The snake visits $D_1, D_2, D_3, \dots$ in sequence, alternating direction by parity:

- **Odd s** — top to bottom: $(1, s-1) \rightarrow (2, s-2) \rightarrow \dots \rightarrow (s, 0)$.
- **Even s** — bottom to top: $(s, 0) \rightarrow (s-1, 1) \rightarrow \dots \rightarrow (1, s-1)$.

A cell (r, c) is **circled** — i.e. assigned a position by f — iff c/r is in lowest terms, equivalently

$$\gcd(c, r) = 1,$$

with the convention $\gcd(0, r) = r$, so $0/r$ is circled only at $r = 1$ (the canonical representative of 0).

The function $f(n)$ returns the rational at the n -th circled cell along the walk; $f^{-1}(q)$ returns the position n at which q first appears.

1.3 Cumulative Count Through Diagonal s

Walking $D_1, D_2, \dots$ and tallying circled cells gives:

s	New circled in D_s	Cumulative
1	1	1
2	1	2
3	2	4
4	2	6
5	4	10
6	2	12
7	6	18
8	4	22
9	6	28
10	4	32
11	10	42
12	4	46
13	12	58

Table 1.1

(Each row's “new circled” count is $\varphi(s)$, Euler's totient — see the remark at the end.)

1.4 Computing $f(26)$

Position **26** falls in D_9 (cumulative through D_8 is **22**; D_9 adds **6**, ending at **28**). D_9 is odd, so walk top to bottom:

Cell (r, c)	Rational	Circled?	Position
(1, 8)	8/1	yes	23
(2, 7)	7/2	yes	24
(3, 6)	$6/3 = 2$	no	—
(4, 5)	5/4	yes	25
(5, 4)	4/5	yes	26

Table 1.2

$$f(26) = \frac{4}{5}$$

1.5 Computing $f(30)$

Position **30** falls in D_{10} (cumulative through D_9 is **28**; D_{10} adds **4**, ending at **32**). D_{10} is even, so walk bottom to top:

Cell (r, c)	Rational	Circled?	Position
(10, 0)	0/10	no	—
(9, 1)	1/9	yes	29
(8, 2)	$2/8 = 1/4$	no	—
(7, 3)	3/7	yes	30

Table 1.3

$$f(30) = \frac{3}{7}$$

1.6 Computing $f^{-1}(3/4)$

The rational $3/4$ sits at cell $(r, c) = (4, 3)$, so $s = 7$. D_7 is odd; cumulative through D_6 is 12, so D_7 starts at position 13.

Cell (r, c)	Rational	Circled?	Position
(1, 6)	6/1	yes	13
(2, 5)	5/2	yes	14
(3, 4)	4/3	yes	15
(4, 3)	3/4	yes	16

Table 1.4

$$f^{-1}(3/4) = 16$$

1.7 Computing $f^{-1}(6/7)$

The rational $6/7$ sits at cell $(r, c) = (7, 6)$, so $s = 13$. D_{13} is odd; cumulative through D_{12} is 46, so D_{13} starts at position 47.

Cell (r, c)	Rational	Circled?	Position
(1, 12)	12/1	yes	47
(2, 11)	11/2	yes	48
(3, 10)	10/3	yes	49
(4, 9)	9/4	yes	50

(5, 8)	8/5	yes	51
(6, 7)	7/6	yes	52
(7, 6)	6/7	yes	53

Table 1.5

$$f^{-1}(6/7) = 53$$

1.8 Summary

$$f(26) = \frac{4}{5}, \quad f(30) = \frac{3}{7}, \quad f^{-1}(3/4) = 16, \quad f^{-1}(6/7) = 53.$$

1.9 Remark — The Totient Pattern

The “new circled” count for D_s is exactly $\varphi(s)$, Euler’s totient. Why: cells in D_s are (r, c) with $r + c = s$, and for any such pair,

$$\gcd(c, r) = \gcd(c, s - c) = \gcd(c, s) = \gcd(s, r),$$

using the Euclidean property $\gcd(a, b) = \gcd(a, b + ka)$ for any integer k . So the circled cells of D_s correspond to indices $r \in \{1, 2, \dots, s\}$ with $\gcd(r, s) = 1$ — precisely $\varphi(s)$ of them.

Cumulating,

$$\text{cumulative through } D_s = \sum_{k=1}^s \varphi(k) = 1, 2, 4, 6, 10, 12, 18, 22, 28, 32, 42, 46, 58, \dots$$

This is the *totient summatory function* $\Phi(s)$, closely related to counting fractions in the Farey sequence. Every non-negative rational appears exactly once along the snake — and that, packaged as $f : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0}$, is the bijection witnessing that $\mathbb{Q}_{\geq 0}$ (and hence $\mathbb{Q}$) is countably infinite.